



MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

COURSE PLAN

Department	:	Biotechnology			
Course Name & code	:	Principles of heat and mass transfer & BIO2254			
Semester & branch	:	IV & Biotechnology			
Name of the faculty	:	M.Ramananda Bhat and Subbalaxmi S			
No of contact hours/week:		L	T	P	C
		3	1	0	4

ASSESSMENT PLAN

Course Outcomes (COs)

<i>At the end of this course, the student should be able to:</i>		No. of Contact Hours	Marks
CO1:	Explain the driving force, various modes of heat transfer such as conduction, convection.	8	20
CO2:	Comprehend the concept of thermal conductivity, heat transfer coefficient, black body radiation and simultaneous heat and mass transfer	9	20
CO3:	Will be able to solve problems based on heat transfer	12	20
CO4:	Comprehend the concept of driving force required for Mass Transfer operations such as simple diffusion, absorption, distillation.	10	20
CO5:	Use, apply Fick's law and comprehend various theories of mass transfer, mass transfer coefficients, NTU, HTU & their significances.	9	20

	Total	48	100
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Components	Quizzes	Sessional Tests	End Semester/ Make-up Examination
Duration	20 to 30 minutes	60 minutes	180 minutes
Weightage	20 % (4 X 5 marks)	30 % (2 X 15 Marks)	50 % (1 X 50 Marks)
Typology of Questions	Understanding/ Comprehension; Application; Analysis; Synthesis; Evaluation	Knowledge/ Recall; Understanding/ Comprehension; Application	Understanding/ Comprehension; Application; Analysis; Synthesis; Evaluation
Pattern	Answer one randomly selected question from the problem sheet (Students can refer their class notes)	MCQ: 10 questions (0.5 marks) Short Answers: 5 questions (2 marks)	Answer all 5 full questions of 10 marks each. Each question may have 2 to 3 parts of 3/4/5/6/7 marks
Schedule	4, 7, 10, and 13 th week of academic calendar	Calendared activity	Calendared activity
Topics Covered	Quiz 1 (L _{x1-x2} & T _{y1-y2}) (CO1,2)	Test 1 (L _{a1-a2} & T _{b1-b2}) (CO1-3)	Comprehensive examination covering full syllabus. Students are expected to answer all questions (CO1-5)
	Quiz 2 (L _{x3-x4} & T _{y3-y4}) (CO2,3)		
	Quiz 3 (L _{x5-x6} & T _{y5-y6}) (CO3,4)	Test 2 (L _{a3-a4} & T _{b3-b4}) (CO3- 5)	
	Quiz 4 (L _{x7-x8} & T _{y7-y8}) (CO4,5)		

Course Plan

L. No.	Topics	Course Outcome Addressed
L0	Introduction to the course	1
L1	Different modes of heat transfer	1
L2	Fourier's law – conduction through plane wall, cylinder	3
T1	Tutorial on Fouriers law	1
L3	Variation of thermal conductivity with temperature	1
L4	Heat flow through multilayer slab.	1
L5	Heat flow through multilayer cylinder and sphere	1
T2	Tutorial on multilayer flow	3

L6	Conduction with heat generation	1
L7	Biological heat production-microbial system	1
L8	Physical mechanism of convection	2
T3	Tutorial on Convection and conduction with heat generation	3
L9	Thermal boundary layer of convection	2
L10	Concept of natural convection over surfaces	2
L11	Forced convection and correlations	2
T4	Tutorial on heat transfer coefficients	3
L12	Dimensional analysis	2
L13	Various dimensionless numbers used in heat transfer	2
L14	Heat exchangers and its classification	3
T5	Tutorial on dimensionless numbers	3
L15	Log mean temperature difference for heat exchangers	3
L16	Heat transfer with phase change	3
L17	Different regimes in boiling.	3
L18	Functioning of evaporators	3
L19	Types of evaporators, boiling point elevation	3
T6	Tutorial on heat transfer with phase change	3
L20	Radiation heat transfer	2
L21	Kirchoff's law	2
L22	Stefan Boltzman's law, application	2
T7	Tutorial on radiation	3
L23	Introduction to mass transfer and different Mass Transfer Operations	4
L24	Concept of driving force	4
L25	Analogies between momentum heat and mass transfer	4
L26	Diffusion mass transfer- Fick's law	4
L27	Molecular and eddy diffusion in gases and liquids.	4
T8	Calculation based on Fick's law	4
T9	Steady state diffusion under stagnant conditions and calculations	4

L28	Steady state diffusion under equimolal flow conditions and calculations	4
L29	Diffusion in solids, its application.	4
L30	Various theories of mass transfer	4
L31	Concept of mass transfer coefficient	5
T10	Problems based on theories of mass transfer	5
L32	Two film theory and equilibrium diagram	5
T11	Calculations based on two film theory	5
T12	Calculations based on two film theory	5
L33	Various dimensionless number in mass transfer	5
L34	HTU and NTU	5
L36	Stages in mass transfer	5
L37	Efficiency in mass transfer	5
L38	Revision	

References:

1. Unit Operations of Chemical Engineering, McCabe & Smith, McGraw Hill Inc, , 1993
2. Mass transfer, R.E. Treybal, McGraw Hill, 1981
3. Fundamentals of Heat and Mass Transfer, Incropera, Dewitt, Wiley Publications
4. Process Heat Transfer, Donald Kern, Mc Graw Hill publications
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Submitted by: M.RAMANANDA BHAT

(Signature of the faculty)

Date: 11-01-2020

Approved by: DR. M. RAMANANDA BHAT

(Signature of HOD)

Date: 11-01-2020

FACULTY MEMBERS TEACHING THE COURSE (IF MULTIPLE SECTIONS EXIST):

FACULTY	SECTION	FACULTY	SECTION
